Name: Aarav Sharma | USN / Roll No: 1PH23CS001

Subject: Engineering Physics - IA 1 | Section: A

Ans 1)

Newton's first law: A body continues in its state of rest or of uniform motion in a straight line unless an external force acts on it. This is also called the law of inertia.

Second law: Rate of change of momentum is directly proportional to the applied force and is in the direction of the force. F = ma.

Third law: For every action there is an equal and opposite reaction. Eg: when a gun is fired the gun recoils backwards.

Ans 2)

Given u = 0, v = 20 m/s, t = 5 s

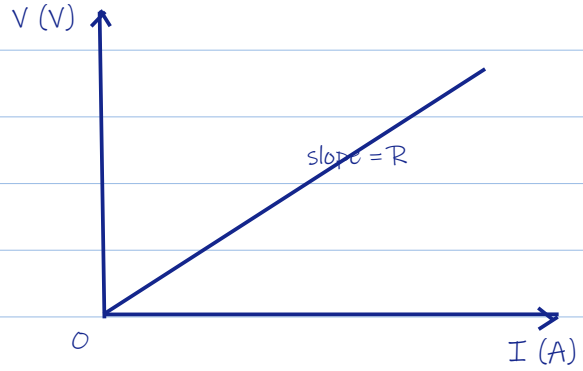
(a) a = (v - u)/t = (20 - 0)/5 = 4 m/s^2

(b) s = ut + 1/2 at^2 = 0 + 1/2 x 4 x 5^2 = 50 m

Distance = 50 m

Ans 3)

Ohm's law: At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends. $V = IR$ where R is the resistance.



The graph is a straight line passing through the origin. Resistance $R = V/I$ = slope of the V-I graph, which is constant for an ohmic conductor.

Ans 4)

KE is the energy due to motion, $KE = 1/2\ mv^2$. PE is the energy due to position (height), $PE = mgh$.

By conservation of energy, $mgh = 1/2\ mv^2$

$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = \sqrt{400} = 20$ m/s